TECHNICAL ASSIGNMENT

AI Research Intern

Binary Change Detection on EO-SAR Image Pairs

1.Abstract :

This project focuses on satellite image-based change detection using deep learning for identifying **structural and environmental changes** between pre-event and post-event imagery. A baseline **U-Net segmentation architecture** was implemented to predict binary change masks from temporally separated satellite images. The input consisted of concatenated pre-event and post-event images, while the output represented pixel-level change localization.

The workflow included dataset analysis, preprocessing, model training, evaluation, threshold tuning, and prediction visualization. Due to severe class imbalance in the dataset, specialized loss weighting techniques were applied to improve sensitivity toward sparse change regions. Model performance was evaluated using Intersection over Union (IoU), Precision, Recall, F1 Score, and confusion matrix analysis.

Experimental results showed that the **model successfully learned coarse change patterns** but **struggled with precise localization** because of highly imbalanced data and sparse target regions. **Threshold tuning improved prediction stability and reduced excessive false positives**. The project demonstrates a complete end-to-end pipeline for satellite change detection and highlights the practical challenges of imbalance handling in remote sensing segmentation tasks.

2.Literature Survey

Paper / Method	Key Idea	Limitation	Relation to Our Work
ChangeFormer	Transformer-based change detection	Computationally expensive	Inspired modern change detection approaches
U-Net	Encoder–decoder segmentation	Struggles with imbalance	Used as baseline architecture
DHFF	EO-SAR feature fusion	Complex optimization	Motivated multimodal fusion
DRNet	Deformable CNN for SAR	SAR-only focus	Highlighted spatial feature extraction
mSwinUNet	Swin Transformer fusion	High resource requirement	Shows future improvement direction

Existing research demonstrates strong performance using **transformer-based** and **multimodal fusion architectures** for satellite change detection. However, many approaches require **significant computational resources and large-scale annotated datasets**. **This work instead focuses** on a **lightweight U-Net baseline** capable of handling EO-SAR change detection with **limited resources**. The study also investigates the impact of class imbalance and threshold tuning on segmentation performance.

3. Methodology

3.1 Overview

The objective of this work is to perform binary change detection using temporally separated satellite imagery. The model receives pre-event and post-event images and predicts a pixel-wise binary mask representing regions of change.

The overall pipeline consists of:

1. Dataset preprocessing and analysis
2. Multimodal image fusion
3. Deep learning-based segmentation
4. Loss optimization with imbalance handling
5. Threshold-based prediction refinement
6. Evaluation using segmentation metrics

3.2 Dataset Preparation

The dataset contains:

- Pre-event satellite images
- Post-event satellite images
- Ground-truth binary change masks

Each sample consists of paired temporal observations and a corresponding segmentation target.

The images were resized to: 256×256

to reduce computational overhead and enable feasible training on limited hardware resources.

The pre-event and post-event RGB images were concatenated channel-wise to create a unified input tensor:

$3+3=6$

resulting in an input tensor of shape:

—> [Batch Size, 6, Height, Width]

3.3 Model Architecture

A baseline U-Net architecture was used for semantic segmentation-based change detection.

The network follows an encoder–decoder structure:

- The encoder extracts hierarchical spatial features.
- The bottleneck captures contextual semantic information.
- The decoder reconstructs pixel-level segmentation masks.

Skip connections were incorporated between encoder and decoder stages to preserve fine-grained spatial information and improve localization of small change regions.

The final output layer predicts a binary segmentation mask representing:

- `1` → change
- `0` → no change

3.4 Training Strategy

The model was trained using supervised learning on paired temporal satellite images.

Training Configuration:

Parameter	Value
Batch Size	2
Image Size	256 × 256
Optimizer	Adam
Learning Type	Binary Segmentation
Epochs	10–35 epochs
Framework	PyTorch

Training performed on GPU-based Google Colab sessions.

3.5 Loss Function

The task exhibited severe class imbalance, where change pixels occupied only a very small portion of the dataset.

To address this, a combined loss function was implemented:

1. Binary Cross Entropy (BCE) Loss
2. Dice Loss

The final loss function was:

- L(total) = L(BCE) + L(Dice)
- **Binary Cross Entropy Loss**

  BCE loss penalizes incorrect pixel-wise predictions.

- **Dice Loss**

  Dice loss improves overlap learning between predicted and target masks and is particularly useful for sparse segmentation tasks.

3.6 Handling Class Imbalance

Dataset analysis revealed that approximately:

- 98% of pixels belonged to the no-change class
- only ~2% represented actual changes

This caused the model to initially predict mostly background regions.

To improve sensitivity toward rare change pixels:

- weighted BCE loss was introduced using `pos_weight`
- prediction thresholds were experimentally tuned

The positive class weight was progressively increased during experimentation to encourage the model to focus more on change regions.

3.7 Threshold Tuning

The raw model outputs were converted to binary masks using sigmoid activation followed by thresholding.

Different thresholds were tested:

- 0.5
- 0.3
- 0.2

Lower thresholds increased recall but produced excessive false positives, while higher thresholds reduced false alarms but missed smaller changes. Threshold tuning was therefore used to balance sensitivity and prediction stability.

3.8 Experimental Iterations

Multiple experiments were conducted throughout development.

Experiment	Observation
Initial Training	Model predicted mostly no-change regions
Increased Epochs	Reduced training loss but limited segmentation improvement
Lower Thresholds	Improved recall but increased false positives
Higher Positive Class Weight	Increased sensitivity to sparse changes
Threshold Refinement	Reduced overprediction and stabilized outputs

3.9 Rationale Behind Design Choices

Design Choice	Reason
U-Net Architecture	Lightweight and effective for segmentation
Channel Concatenation	Simple EO-SAR temporal fusion
Dice + BCE Loss	Better sparse-region segmentation
Weighted BCE	Address severe imbalance
Threshold Tuning	Control false positives vs recall
Image Resizing	Reduce memory and training cost

4. Results

4.1 Quantitative Evaluation

The trained U-Net model was evaluated on the validation and test datasets using standard segmentation metrics for the change class (`label = 1`).

The evaluation metrics included:

- Intersection over Union (IoU)
- Precision
- Recall
- F1 Score
- Confusion Matrix

A prediction threshold of: **Threshold=0.5**

was selected after experimentation to reduce excessive false positives generated at lower thresholds.

4.2 Validation Set Results

The trained U-Net model was evaluated on the validation split using standard segmentation metrics for the change class (`label = 1`). The evaluation was performed after applying weighted loss balancing and threshold tuning.

Metric	Score / Value
IoU	0.0127
Precision	0.0231
Recall	0.0583
F1 Score	0.0229

Confusion Matrix (Validation Set)

Actual \ Predicted	Predicted No Change	Predicted Change
Actual No Change	16,912,570	4,480,376
Actual Change	397,509	98,569

4.3 Test Set Results

Metric	Value / Count
IoU	0.0072
Precision	0.0081
Recall	0.1184
F1 Score	0.0137
True Negatives	3,884,505
False Positives	1,120,069
False Negatives	30,610
True Positives	11,088

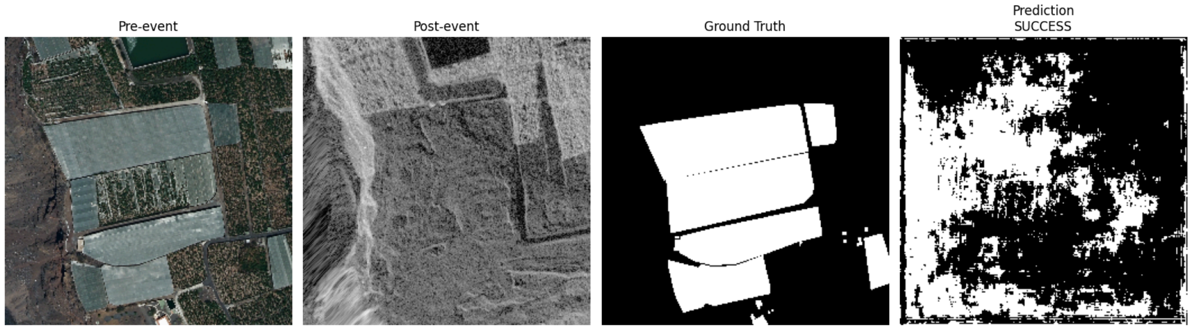
4.4 Qualitative Visualizations

The following visualizations compare:

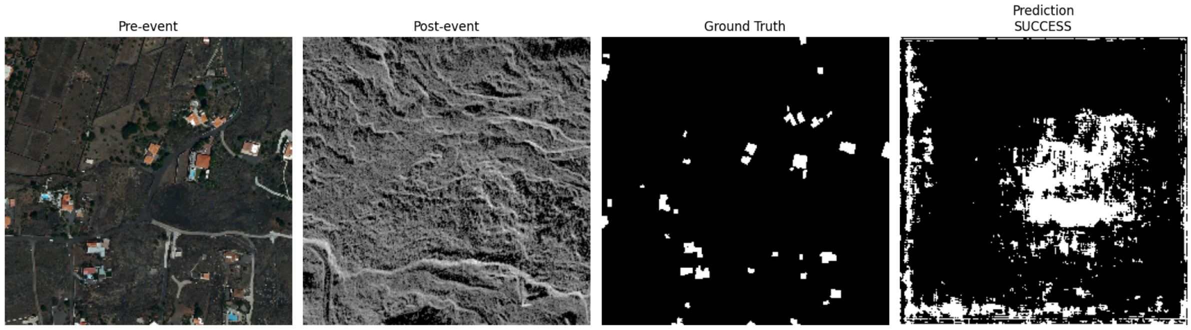
- Pre-event image
- Post-event image
- Ground-truth mask
- Predicted mask

These examples include both successful detections and failure cases.

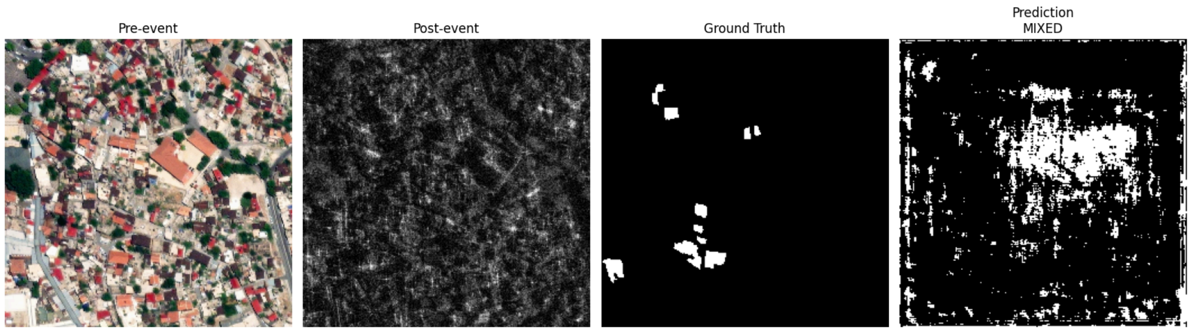
Example 1:(Success)



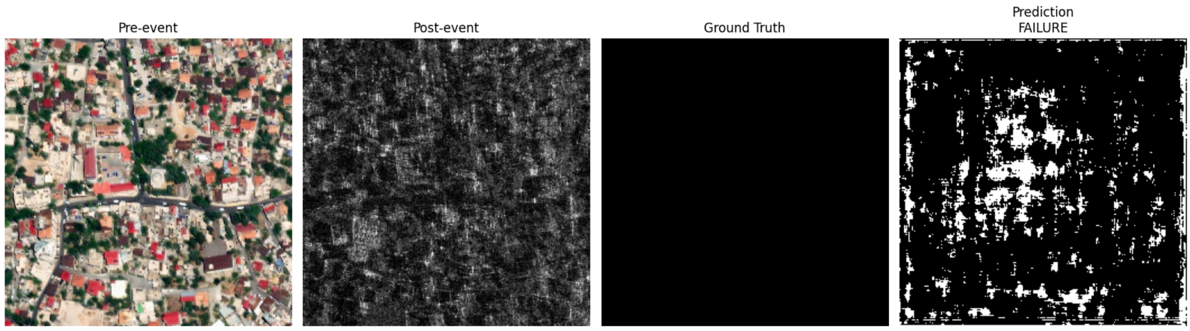
Example 2:(Success)



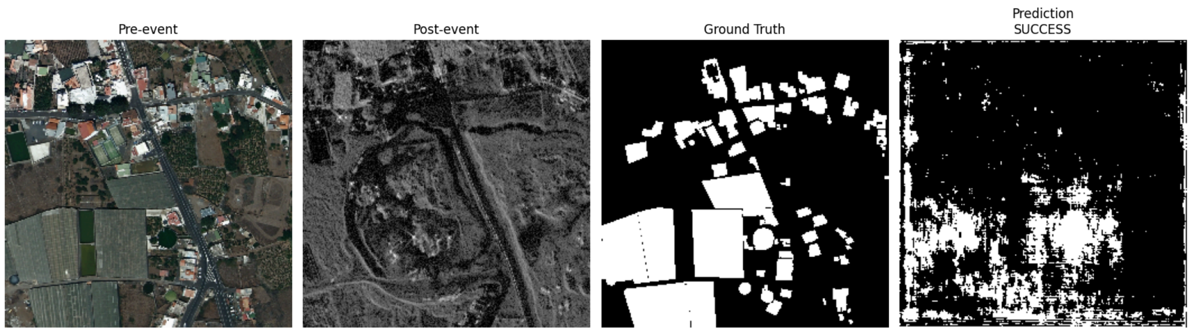
Example 3:(Mixed)



Example 4:(Failure)



Example 5(Success):



4.5 Result Analysis

The experiments demonstrate that the model successfully learned coarse temporal change patterns from EO-SAR imagery but struggled with fine-grained localization of sparse damage regions.

The major observations are:

1. Severe class imbalance significantly affected segmentation quality.
2. Lower thresholds improved recall but produced excessive false positives.
3. Higher thresholds reduced false positives but missed subtle changes.
4. The baseline U-Net architecture showed limited capability in modeling complex multimodal spatial relationships.

The confusion matrix indicates that the model achieved moderate sensitivity toward change regions but suffered from substantial false-positive predictions, leading to low precision and IoU scores.

Despite these limitations, the model successfully established a complete end-to-end satellite change detection pipeline including:

- preprocessing,
- multimodal fusion,
- segmentation training,
- threshold tuning,
- evaluation,
- and qualitative analysis.

5. Future Work

The current work establishes a baseline EO-SAR change detection pipeline using a U-Net segmentation model. While the model learned coarse change patterns, performance was **limited by severe class imbalance, sparse target regions, and high false-positive predictions.** Future work would focus on improving multimodal fusion, segmentation accuracy, and robustness.

Key Future Directions

Area	Planned Improvement	Motivation
Architecture	Attention U-Net / Siamese U-Net	Improve temporal feature comparison and localization
Transformer Models	ChangeFormer / Swin Transformer	Capture long-range spatial and multimodal dependencies
Loss Functions	Focal Loss and Dice-Focal Loss	Better handling of class imbalance
EO-SAR Fusion	Feature-level and attention-based fusion	Improve interaction between optical and SAR modalities
Training Strategy	Patch-based and multi-scale training	Preserve fine structural details
Data Augmentation	Rotation, cropping, SAR noise augmentation	Improve generalization and robustness
Post-processing	Morphological filtering and CRFs	Reduce noisy false-positive regions
Infrastructure	GPU scaling and mixed-precision training	Enable larger and faster experimentation

The current baseline demonstrates a complete end-to-end EO-SAR change detection workflow but also highlights the complexity of multimodal remote sensing segmentation tasks. Future work would prioritize:

- stronger multimodal fusion architectures,
- imbalance-aware optimization,
- high-resolution learning,
- and transformer-based contextual modeling.

These directions have strong potential to significantly improve localization accuracy, reduce false positives, and enable practical large-scale satellite change detection systems.

6. Conclusion

This work presented a baseline deep learning pipeline for EO-SAR satellite image change detection using a U-Net segmentation architecture. The model was trained on paired pre-event and post-event imagery to generate binary change masks representing structural and environmental changes.

The project successfully implemented the complete workflow including:

- dataset preprocessing,
- multimodal image fusion,
- segmentation model training,
- imbalance-aware loss optimization,
- threshold tuning,
- quantitative evaluation,
- and qualitative visualization.

Experimental results showed that the model was able to learn coarse temporal change patterns and detect portions of true change regions. However, performance remained limited due to severe class imbalance, sparse target regions, and high false-positive predictions. The baseline **U-Net architecture also struggled with precise localization of fine-grained changes in heterogeneous EO-SAR imagery.**

The experiments highlighted the strong impact of:

- class imbalance handling,
- threshold selection,
- and multimodal representation learning
  on satellite change detection performance.

Overall, this work establishes a functional end-to-end baseline for EO-SAR change detection while clearly identifying the limitations of simple segmentation architectures. Future improvements using transformer-based fusion models, advanced imbalance-aware losses, and higher-resolution training strategies are expected to significantly improve detection quality and robustness.

7. Time and Resource Log

7.1 Time Breakdown

Activity	Approximate Time Spent
Dataset Exploration and Analysis	3-4 hours
Literature Reading and Research	4–6 hours
Data Preprocessing and Debugging	7–8 hours
Model Implementation	4–5 hours
Training and Hyperparameter Tuning	12–13 hours
Evaluation and Visualization	2–3 hours
Report Writing and Documentation	3–4 hours
Total Estimated Time	**35–43 hours**

7.2 Training Environment

Resource	Details
Local Machine	MacBook Air
Cloud Platform	Google Colab
GPU Availability	Limited due to usage restrictions

GPU Model Used	NVIDIA T4 (when available in Colab)
GPU VRAM	16 GB
Number of GPUs	1
Deep Learning Framework	PyTorch

7.3 Training Runtime

Parameters	Approximate Value
Image Resolution Used	256 × 256
Batch Size	2
Training Time per Epoch	~8–15 minutes (GPU) / significantly higher on CPU
Total Epochs Trained	Multiple runs between 10–35 epochs

Parameters	Approximate Value
Total Wall-Clock Training Time	~6–10 hours cumulative

7.4 Constraints Faced

Several practical constraints influenced the implementation and experimentation strategy:

- Limited and intermittent GPU availability on Google Colab
- Restricted local hardware resources for high-resolution training
- Severe dataset imbalance causing unstable predictions
- Large dataset size leading to storage and transfer limitations
- Long training times for high-resolution satellite imagery

These constraints influenced multiple design decisions, including:

- resizing images to 256 × 256,
- using a lightweight baseline U-Net architecture,
- limiting batch size,
- and prioritizing computationally feasible experimentation over large-scale transformer training.

Despite these limitations, the project successfully established a complete EO-SAR change detection pipeline including preprocessing, training, evaluation, visualization, and analysis.